

Advanced (Habanero)

- Q1.** What is the purpose of a scatter plot?
(A) To show the relationship between two variables
(B) To display a single data point
(C) To create a line graph
(D) To make predictions
(E) To identify trends
- Q2.** Which type of correlation is shown by the equation $y = 2x$?
(A) Negative
(B) None
(C) Positive
(D) Linear
(E) Exponential
- Q3.** What is the line of best fit used for?
(A) To create a scatter plot
(B) To identify correlation
(C) To make predictions based on data trends
(D) To display a single data point
(E) To show the relationship between two variables
- Q4.** If a scatter plot shows a negative correlation, what happens to y as x increases?
(A) y increases
(B) y decreases
(C) y remains the same
(D) y becomes undefined
(E) y becomes zero
- Q5.** What type of relationship is shown by the equation $y = -3x$?
(A) Positive linear
(B) Negative linear
(C) Non-linear
(D) Exponential
(E) Logarithmic
- Q6.** How can you determine the strength of a correlation?
(A) By looking at the slope of the line of best fit
(B) By examining the y -intercept
(C) By calculating the coefficient of determination (R^2)
(D) By creating a new scatter plot
(E) By using a different type of graph
- Q7.** If a line of best fit has a slope of -2 and a y -intercept of 5 , what is the equation of the line?
(A) $y = 2x + 5$
(B) $y = -2x + 5$
(C) $y = 2x - 5$
(D) $y = -2x - 5$
(E) $y = 5x + 2$
- Q8.** What is the purpose of R^2 in a linear regression?
(A) To calculate the slope of the line
(B) To determine the y -intercept
(C) To measure the strength of the correlation
(D) To create a new scatter plot
(E) To identify the type of relationship

Open Ended Questions (FRQ)

- Q9.** Given the data points $(1, 2)$, $(2, 3)$, $(3, 5)$, and $(4, 7)$, create a scatter plot and draw the line of best fit. Use the line to predict the value of y when $x = 5$.
- Q10.** Explain how to determine if a correlation is positive, negative, or none, and provide an example of each type of correlation. Be sure to include a scatter plot and the equation of the line of best fit for each example.

Chef's Tips

- Check for understanding of key concepts before moving on to new material
- Encourage students to use technology to create scatter plots and lines of best fit
- Emphasize the importance of interpreting the results in context